

RMO(INDIA) – 2014

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Let ABC be an acute-angled triangle and $\angle ABC$ is the largest angle of the triangle. Let R be its circumcentre. Suppose the circumcircle of triangle ARB cuts AC again in X . Prove that RX is perpendicular to BC .
 2. Find all real numbers x and y such that $x^2 + 2y^2 + \frac{1}{2} \leq x(2y + 1)$.
 3. Prove that there does not exist any positive integer $n < 2310$ such that $n(2310 - n)$ is a multiple of 2310.
 4. Find all positive real numbers x, y, z such that
$$2x - 2y + \frac{1}{z} = \frac{1}{2014}, \quad 2y - 2z + \frac{1}{x} = \frac{1}{2014}, \quad 2z - 2x + \frac{1}{y} = \frac{1}{2014}.$$
 5. Let ABC be a triangle. Let X be on the segment BC such that $AB = AX$. Let AX meet the circum circle Γ of triangle ABC again at D . Show that the circum centre of $\triangle BDX$ lies on Γ .
 6. For any natural number n , let $S(n)$ denote the sum of the digits of n . Find the number of all 3-digit numbers n such that $S(S(n)) = 2$.

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